INTEL AI FOR FUTURE WORKFORCE BOOTCAMP 2026 SUBMISSION TRACK: AGENTIC AI

DODAI OS: Executive Summary & Project Abstract

Your Local Autonomous Linux Intelligence • Agentic Linux AI Assistant

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< 3.8 ms	100% Offline	7 Agents	135 / 135
DIRECT FAST-PATH LATENCY	LOCAL PRIVACY & SOVEREIGNTY	COORDINATED DAG CORE	TESTS PASSED (100%)

> 💡 **Strategic Vision & Core Concept**
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> DODAI OS is a local-first, verified autonomous AI operating assistant engineered for Linux personal computers. While standard desktop assistants operate as passive chatbots that respond with text, DODAI OS functions as an active operating system companion that understands user objectives, constructs multi-step Directed Acyclic Graph (DAG) execution plans, orchestrates specialized agents, executes real system and development tools, independently verifies ground-truth outcomes, and self-heals through bounded retries.

1. Executive Summary (185 Words)

DODAI OS transforms the Linux desktop into an autonomous, agentic operating workplace. While traditional desktop assistants function merely as passive conversational wrappers that answer prompts with text, DODAI OS operates as an active, permission-aware multi-agent operating system. It possesses the capability to understand high-level user intent, formulate structured multi-step plans as Directed Acyclic Graphs (DAGs), dynamically delegate tasks across specialized autonomous agents, execute system and development tools in sandboxed workspaces, independently verify physical ground-truth results, self-heal through bounded retries, and retain memory across a 7-day rolling window.

DODAI OS uniquely solves the tension between instant responsiveness and deep reasoning through a **Dual-Tier Coexistence Architecture**: Tier 1 provides sub-5ms local DBus and regex dispatch for instant desktop controls without GPU overhead; Tier 2 activates a hierarchical team of 7 specialized agents (Supervisor, Planner, Linux Intelligence, Developer, Research, Tool Execution, Verifier, and Recovery) for complex multi-step workflows. All operations strictly obey a 4-Tier Security Doctrine ensuring zero unauthorized root modifications and strict human-in-the-loop approval gating, making DODAI OS a private, robust, edge-first Agentic AI platform.

2. Project Abstract (310 Words)

Personal computing in the modern knowledge economy remains fundamentally manual. Knowledge workers, developers, and researchers spend considerable cognitive effort switching contexts, diagnosing system health, organizing workspaces, compiling research, scaffolding software projects, and verifying computational workflows. Existing artificial intelligence assistants remain confined to text generation; they lack operating system integration, cannot inspect system states, fail to verify whether their actions succeeded, and rely entirely on external cloud infrastructure that compromises user data privacy.

DODAI OS solves this paradigm bottleneck by introducing an autonomous operating layer tailored for Fedora Linux (KDE Plasma 6 on Wayland). Built upon a local-first philosophy, DODAI OS integrates state-of-the-art quantized neural foundation models (including an ultra-lite 370M LoRA-tuned local brain and multi-modal models) alongside an open-architecture cloud AI bridge. The platform features an autonomous hierarchical multi-agent framework: when a user gives a high-level goal, the Supervisor and Planner Agents construct a Directed Acyclic Graph (DAG) with explicit task dependencies, delegating execution across dedicated Linux Intelligence, Developer, Research, and Tool Execution agents interfacing with 105 native tools.

Crucially, DODAI OS enforces an architectural separation between execution and certification. An independent Verifier Agent evaluates ground-truth physical artifacts (file outputs, process states, stderr streams, unit test pass rates) before declaring task completion. If discrepancies arise, a bounded Recovery Agent analyzes errors and replans execution without infinite loops. Operating alongside a sub-5ms local fast-path for hardware window and media management, continuous acoustic wake recognition via Vosk, Faster-Whisper int8 STT, Piper neural speech synthesis, and 7-day rolling persistent thought retention, DODAI OS establishes a new standard for private, resilient, and verified Agentic AI operating systems on Linux personal computing hardware.

3. Short Description for Google Submission Form

DODAI OS is an autonomous, local-first Agentic AI desktop operating assistant for Linux. Unlike conventional chatbots that produce text, DODAI OS understands complex natural-language goals, decomposes them into Directed Acyclic Graph (DAG) execution plans, and orchestrates a team of specialized agents (Linux Intel, Developer, Research, Tool Execution) to interact with real system tools, scaffold software, execute shell diagnostics, and compile publication-grade reports. It features an independent Verifier Agent that inspects physical filesystem and process ground truth, a bounded self-healing Recovery Agent, a sub-5ms fast path for window and hardware control, a 1.5GB SSD memory engine with 7-day rolling retention, and strict 4-Tier permission containment—delivering fully verified, private desktop autonomy.

4. Five Key Innovations

Innovation	Technical Implementation	Operational Advantage
1. Dual-Tier Coexistence Architecture	Sub-5ms direct KWin DBus & regex fast-path paired seamlessly with an autonomous multi-agent DAG engine.	Zero latency delay on simple tasks (window control, volume) while enabling deep reasoning for complex goals.
2. Independent Verification Gate	Architectural separation where an independent Verifier Agent validates exit codes, physical files, and test results.	Eliminates LLM hallucination and self-certification of incomplete or failed tasks.
3. Bounded Self-Healing Recovery	Runtime error capture triggering root-cause analysis, alternative strategy synthesis, and max 2 bounded retries.	Prevents infinite loops and plan drift while autonomously correcting recoverable errors.
4. Ultra-Lite Local Brain (`nano-brain`)	SmolLM2-360M LoRA fine-tuned on CPU AVX2 FMA execution units (<150MB RAM, sub-350ms generation).	Guarantees sub-second local inference and leaves 98% of system RAM free for actual applications.
5. 7-Day Rolling High-Thought Memory	168-hour rolling memory pool in a 1.5GB NVMe ring buffer with autonomous weekly synthesis (.json & .md).	Preserves historical context, past plans, and user preferences across reboots.

5. Five Important Agentic Capabilities

- **Autonomous Goal Decomposition & DAG Planning:** Decomposes complex multi-faceted objectives into Directed Acyclic Graphs with task dependency tracking.
- **Linux System Intelligence & Process Telemetry:** Directly inspects `/proc`, system load, memory, disk vitals, and listening network sockets without manual user bash commands.
- **Autonomous Software Engineering:** Scaffolds complete multi-file project directories, writes implementation code, and executes automated pytest suites in sandbox.
- **Multi-Facet Web Research & Document Synthesis:** Synthesizes real-world research into publication-grade PDF and Word (.docx) documents with dynamic charts and comparison tables.
- **4-Tier Security & Sandbox Isolation:** Strictly enforces Level 1 (ReadOnly) to Level 4 (Forbidden) boundaries protecting operating system kernel integrity.

6. Why DODAI OS Qualifies as an Agentic AI Project

DODAI OS satisfies every foundational pillar of genuine Agentic AI: (1) *Autonomous Perception* via continuous acoustic listening and operating system state monitoring; (2) *Deliberation* via the Supervisor and Planner Agents that formulate structured multi-step DAG plans; (3) *Delegation* to specialized agents with distinct role boundaries; (4) *Action* through direct execution of 105 Linux desktop tools; (5) *Evaluation* via an independent Verifier Agent checking physical ground truth; (6) *Self-Healing* through bounded recovery replanning; and (7) *Retention* across a 7-day rolling memory pool.

7. Real-World Problem Solved

Eliminates the cognitive friction and manual fragmentation of Linux desktop productivity. Modern engineers and knowledge workers waste 20-30% of their workday on repetitive operations: diagnosing memory leaks, checking network ports, organizing files, converting research into structured reports, scaffolding repetitive boilerplate, and toggling windows. DODAI OS acts as a 24/7 private, local autonomous partner that executes these multi-step workflows end-to-end with verified ground truth—saving hours daily while keeping confidential data 100% on the user's machine.